February 27, 2025

Workstream Status - Starch

International Paper + DS Smith

Substreams

- OME
- Thermal/Steam
- **Strength/Starch (+Overweight)**
- Next Steps
- Appendix

Considerations made for the analysis of the data

- All chemicals used for dry strength considered in the cost, but specific data and conclusions about the efficiency of the surface starch are included.
- Machines with no surface starch (DUE1, VIANA) or very special products (DUE2, KS3) are not included in the study
- Starch consumption taken from Wedge when available, otherwise taken from the BOM.
- All data related to consumption in kg dry starch per gross ton of paper.
- NEW:
  - Madrid detailed analysis included after receiving data per Jumbo.
  - Steam savings due to reduction of starch (surface or cationic) are included
  - Reduction in savings due to the cost of fibre to compensate the reduced starch is included

Strength/Starch: Quartile ranking

Concept	Parameter	AB-01	AL-03	BL-02	KS-04	KS-06	LC-02	MAD	RN-05	WH-01	ZA-01
VARIABILITY between JUMBOS	Weighted average STDev SCT GM	3	1	3	2	4	1	4	2	2	3
VARIABILITY MD	Evaluation in separate file	4		1	1	2	3		2	3	4
MD/CD	Weighted avg MD/CD	2	4	3	4	2	2	1	3	1	3
PRODUCT RANGE	SCT GM MAX-MIN	4	1	1	2	4	2	2	3	3	3
STARCH EFFICIENCY	Slope SCD GM vs Starch	3	2	2	3	4	2	3	1	4	1
OVER QUALITY	Avg SCT GM - Min SCT GM	2		1	4	2	3	4	1	3	2
DRY STRENGTH COST	Cost of each SCT GM index unit	4	2	1	3	3	1	2	3	4	2
OVERALL STRENGHT OPPORTUNITIES		**4**	**2**	**1**	**3**	**4**	**1**	**3**	**2**	**3**	**2**

- **Strength potentials in all mills except Witzenhausen and Kemsley PM-6**
- Total Strength scoring from savings criteria only

Strength related potential savings – Up to $5m

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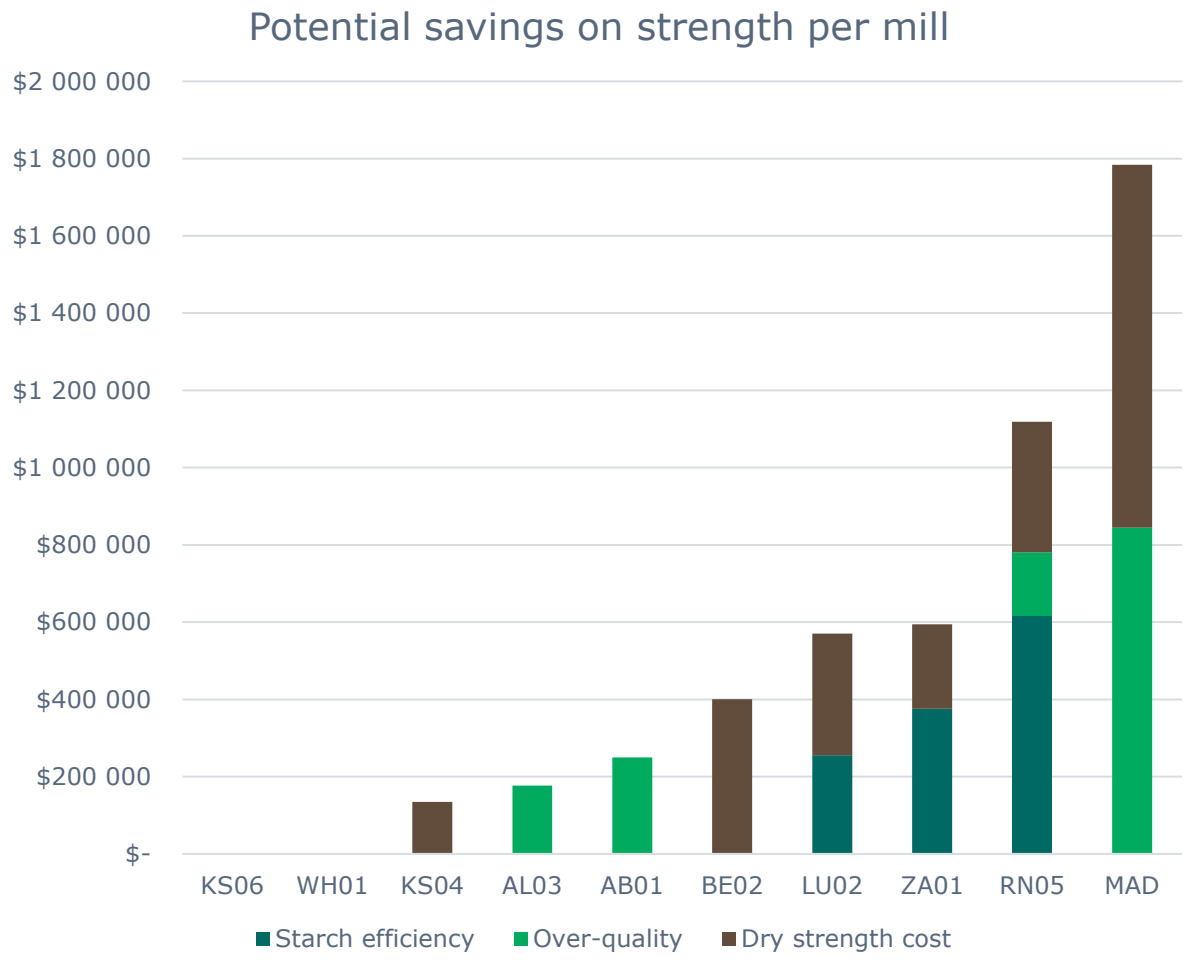


- Comments:
  - Included extra saving in cooking steam
  - Reduced by the cost of fibre to replace reduced chemicals
- Main actions:
  - Cargill audit at specific sites.
  - Improve starch kitchen and implement BAT.
  - Reduce excess dry strength chemicals, remove when possible.
  - Avoid over-quality once variability and measurement bias is solved.

Reference prices:
400 €/Tn of Surface starch,
300 €/Tn Lignosulfonate,
750 €/Tn Cationic Starch
2000 €/Tn Xelorex
3750 €/Tn Hercobond (APAM)
2660 €/Tn CMC

Main ideas to achieve Strength savings (Low CAPEX)

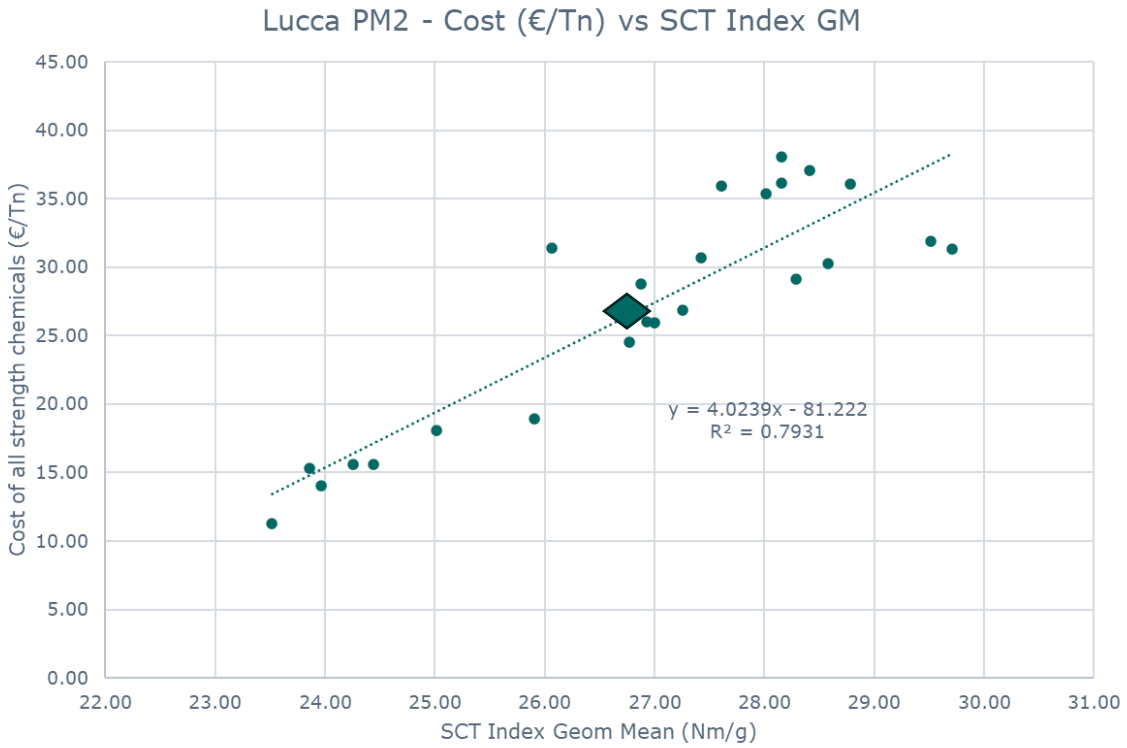
preliminary

Full list submitted separately to Paul J O'Bernier

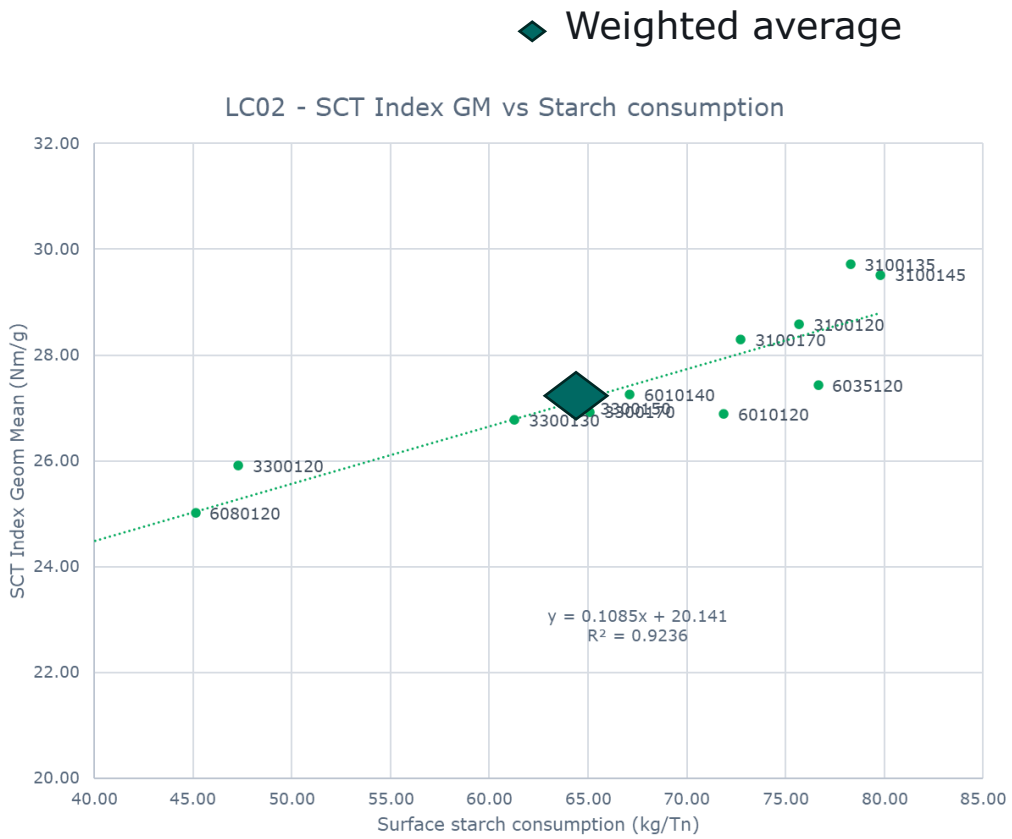
- All EMEA mills fully on board with the Round Robin
- Advanced Process Control implementation to reduce variability and overprocessing
- Cooked starch polymer distribution assessment (Cargill audit)
- Dry strength additives reduction
- Evaluate feasibility of longer cycles (2 week cycles or longer rather than one week cycles)
- Lab conditioning improvement
- MD controls tuned/optimized
- MD/CD ratio optimization (starting by measuring always SCT MD in all the grades)
- Starch cooking as per BAT including installing proper controls and measurements
- Wedge installation and Golden Batch deployment

* Additional request through Guy Lacey prior to Day 1 still pending, but not urgent for this exercise.

Starch saving calculations: Data available per machine



Dry Strength cost per SCT Index unit: Surface starch + Lignosulfonate + Cationic Starch + CPVAm (Xelorex) + CMC



Starch effectiveness = SCT Index Geom Mean increase per Kg/Tn of surface starch added

Starch saving calculations: Data available per machine

- For each machine we evaluate:

Over-quality:

Weighted average SCT CD and GM Index vs Minimum required values according to the product specifications

Variability between Jumbos of same product

Weighted variability in SCT CD and GM between Jumbos (+/- 1 standard deviation considered)

Variability in MD direction

Variations (pope scanner) in Grammage, dry weight and moisture

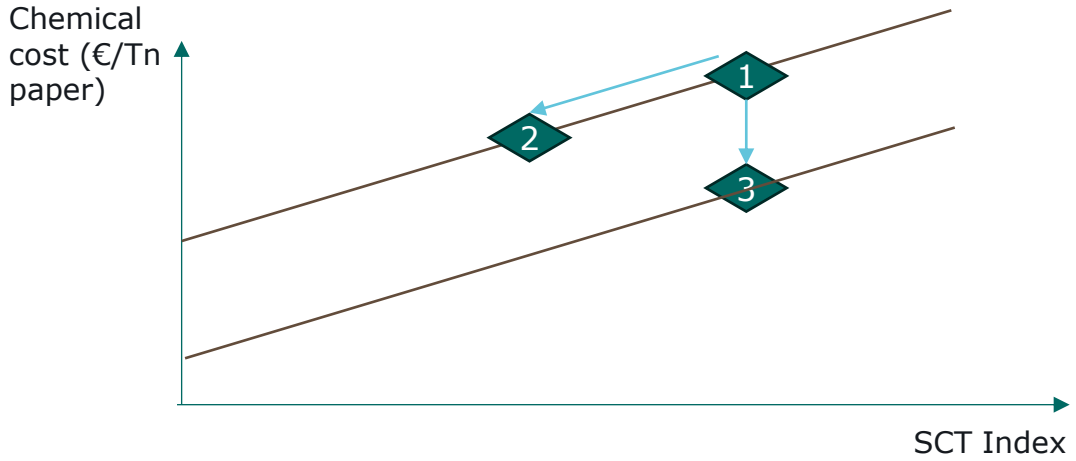
Overweight:

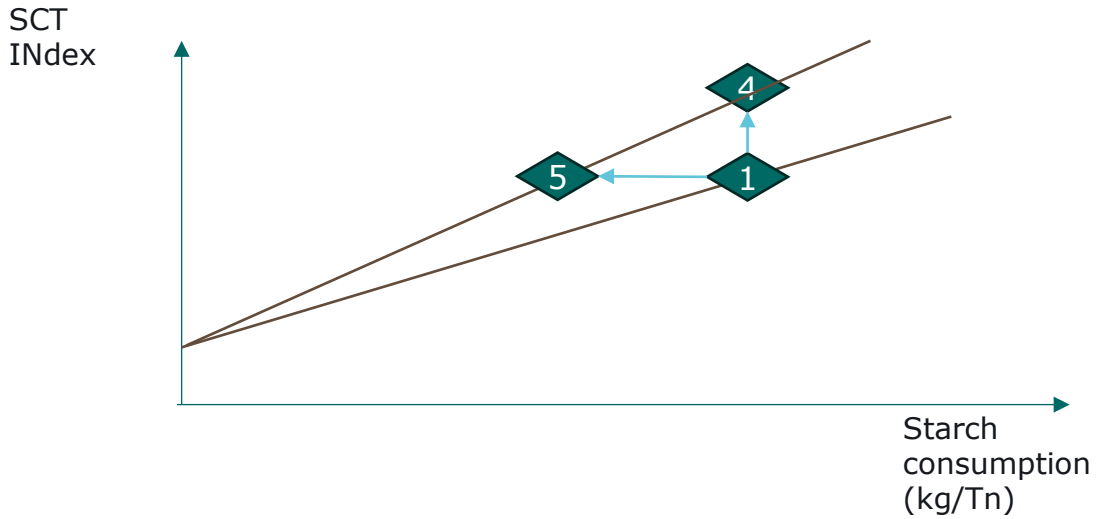
Actual vs nominal grammage (separate calculation)

Fibre orientation:

Weighted SCT MD/SCT CD per product (except for Madrid, Alcolea and Duenas)

Calculating savings in Dry strength chemicals





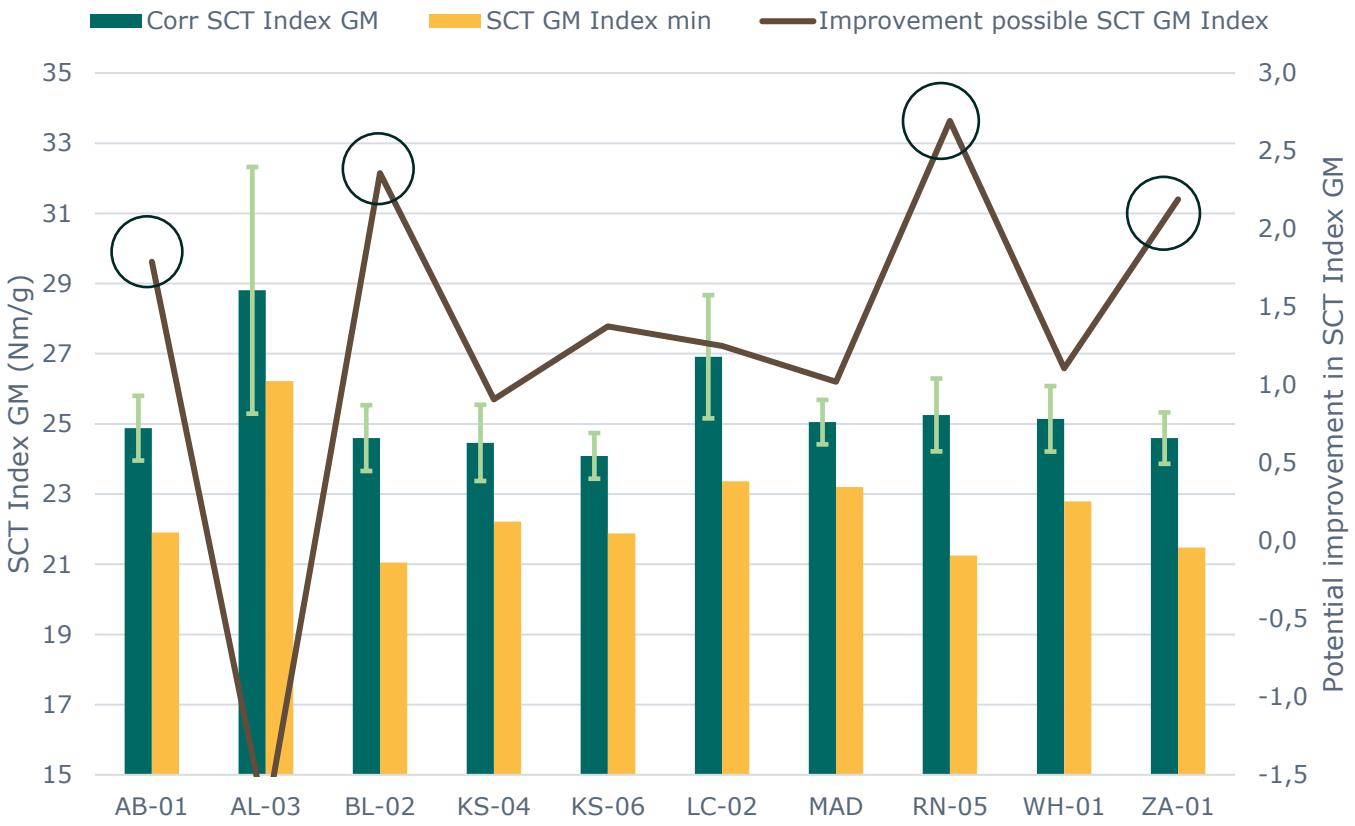
1 → 2: Savings due to over-quality, reduction in cost due to reduced SCT index

3 → 1: Savings due to excess chemical consumption for some grades (example not needed lignosulfonate or cationic starch)

1 → 4: Improved starch efficiency (better cooking, better application), change of the slope that leads to improved product (less need of more expensive chemicals)

1 → 5: Improved starch efficiency that leads to direct surface starch dose reduction

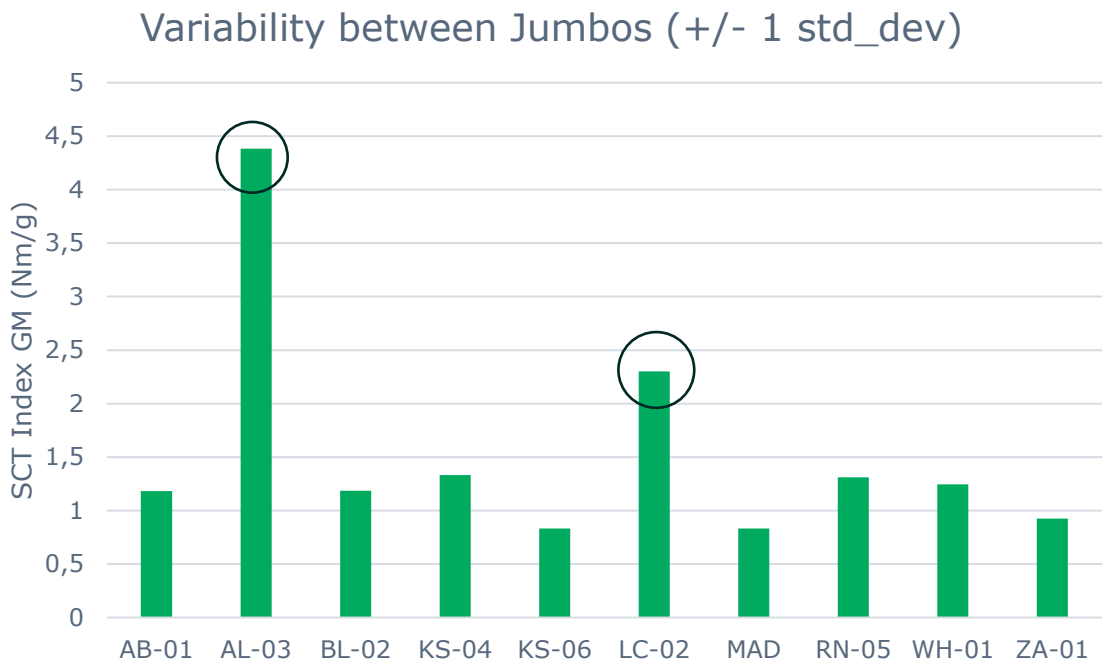
Over-quality

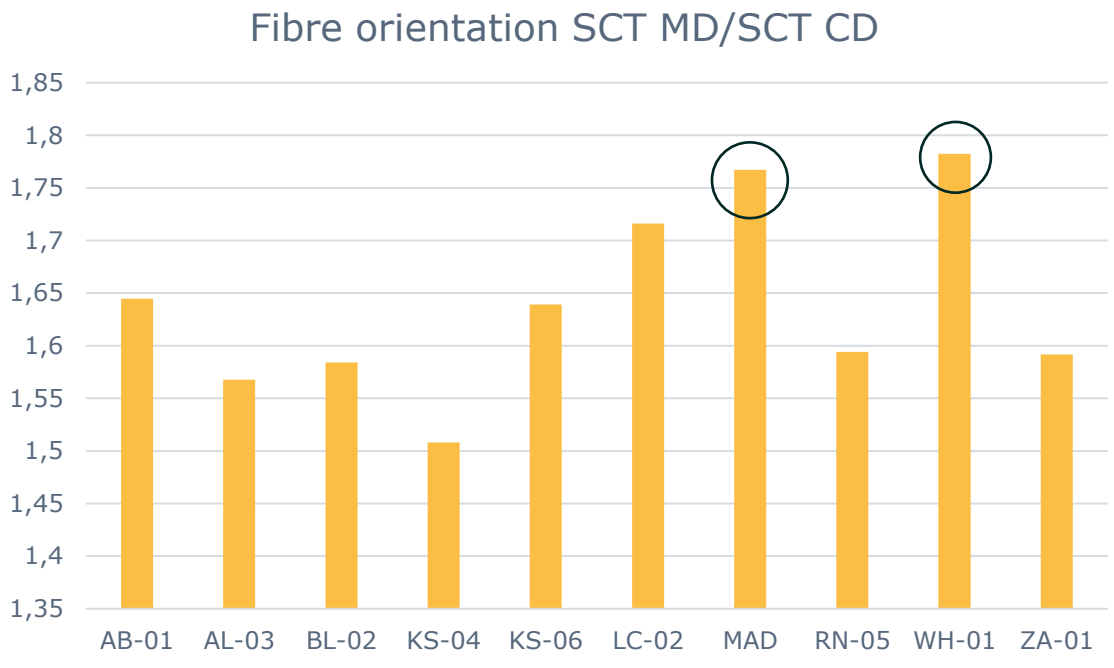


Solving over-quality:

- 1st step is reduce the variability (between Jumbos, inside the jumbo): AL-03, LU-02
- 2nd step is reducing the error in the measurements (i.e: Alcolea)
- 3rd step is improving the MD/CD ratio to the required level (i.e: Madrid)
- 4th step is reducing the SCT set-point in the machines closer to the real demand

Variability between Jumbos and MD/CD

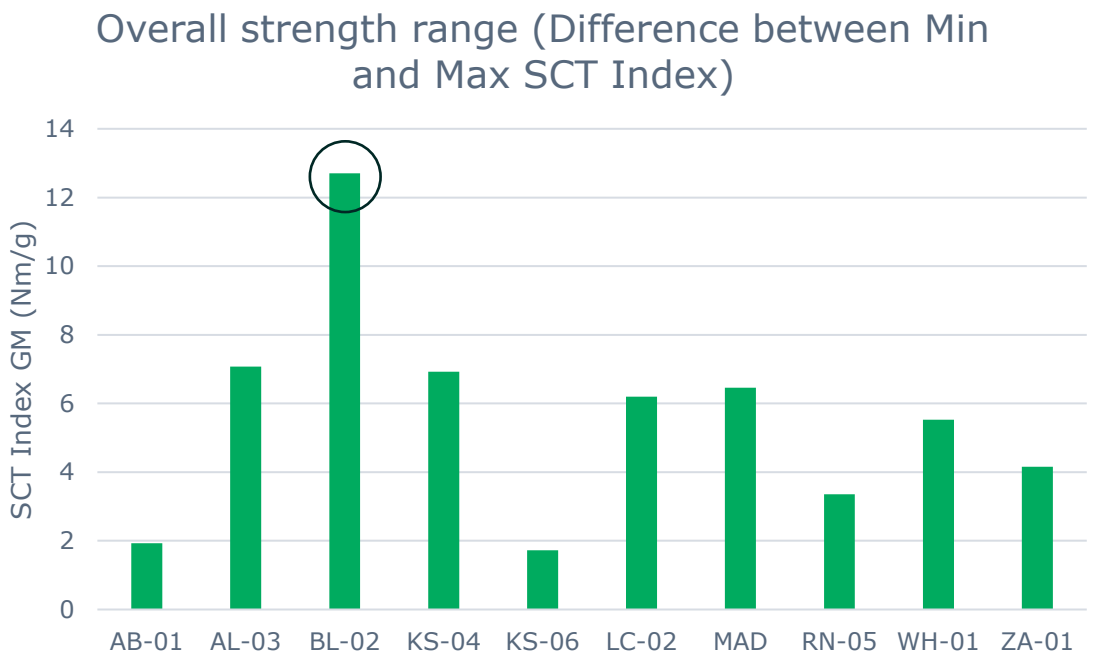


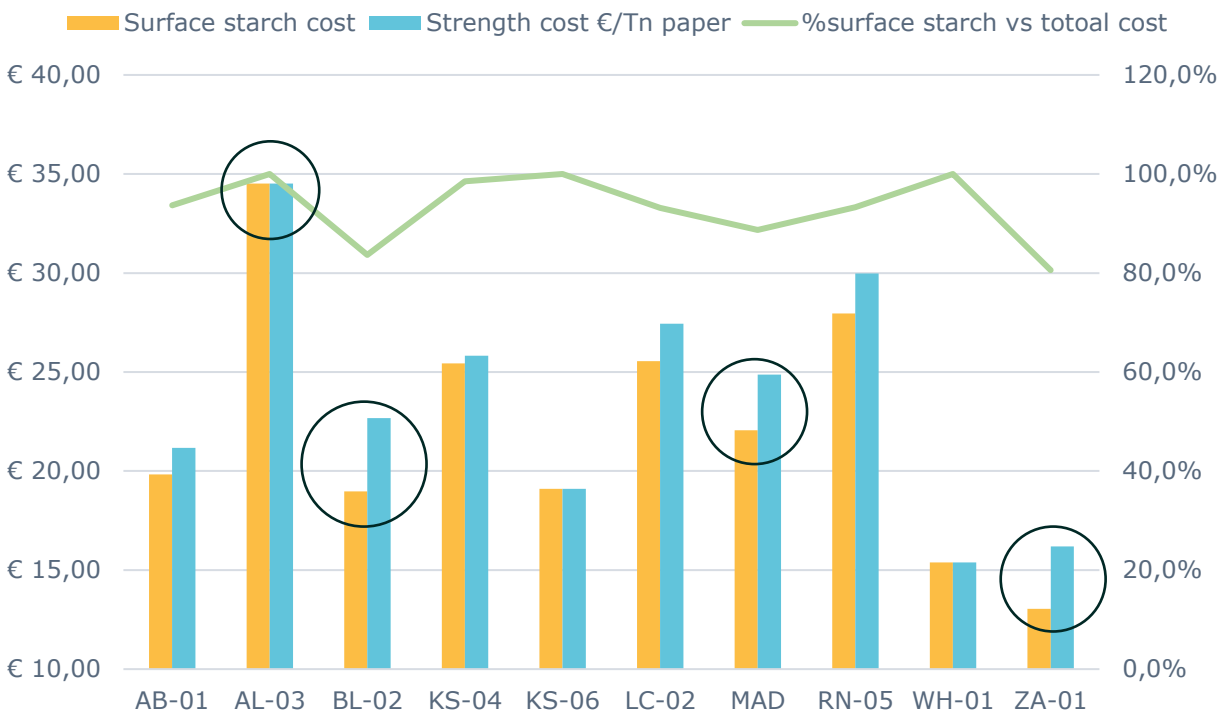


Actions:
Improve measurement and on-line control
Check base paper variability

Actions:
Evaluate strategies per grade to understand and improve the MD/CD (Objective 1.6?)

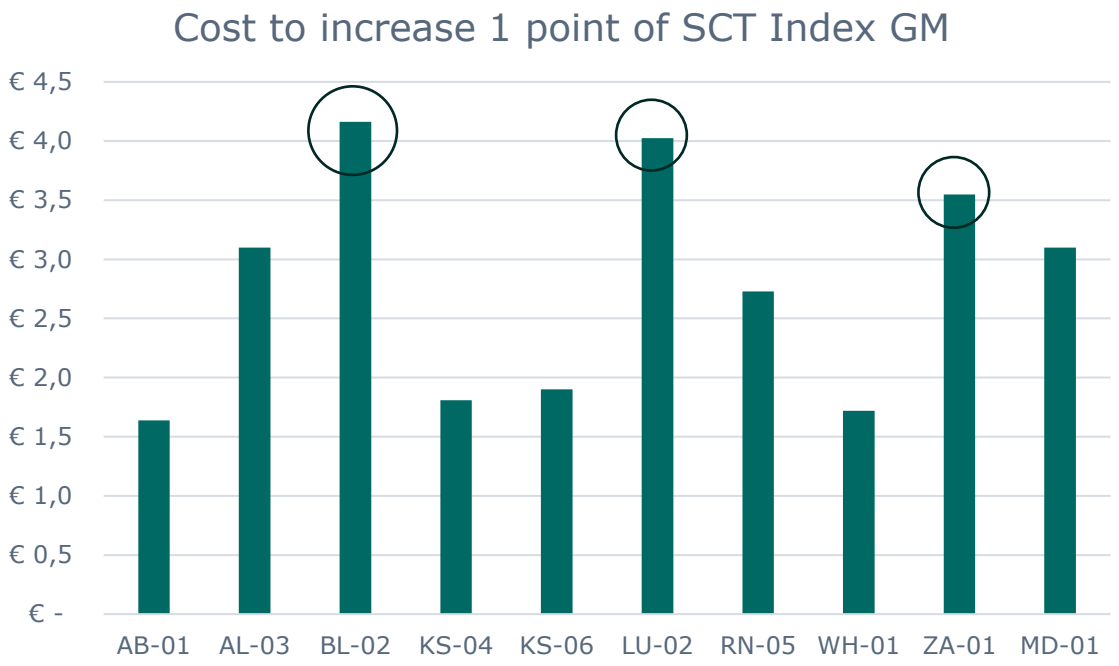
Product range and total strength cost



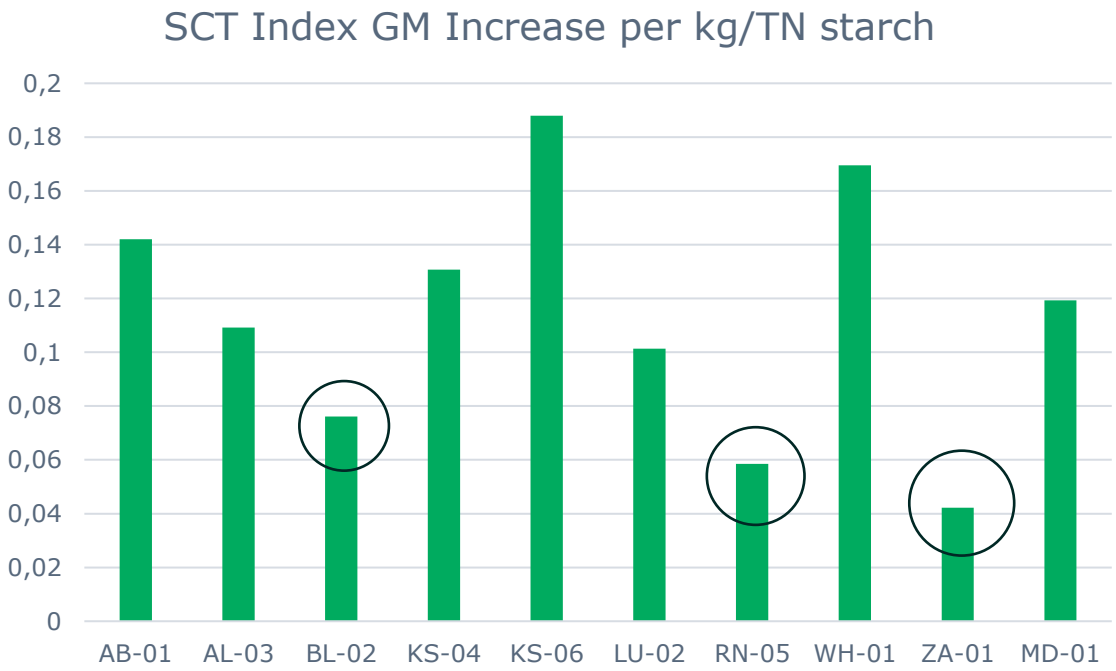


Actions: Evaluate “Sweet spot” in each machine and optimize product range

Strength increase cost and surface starch efficiency



Actions: Remove excess of other chemicals



Actions: Improve starch cooking and application

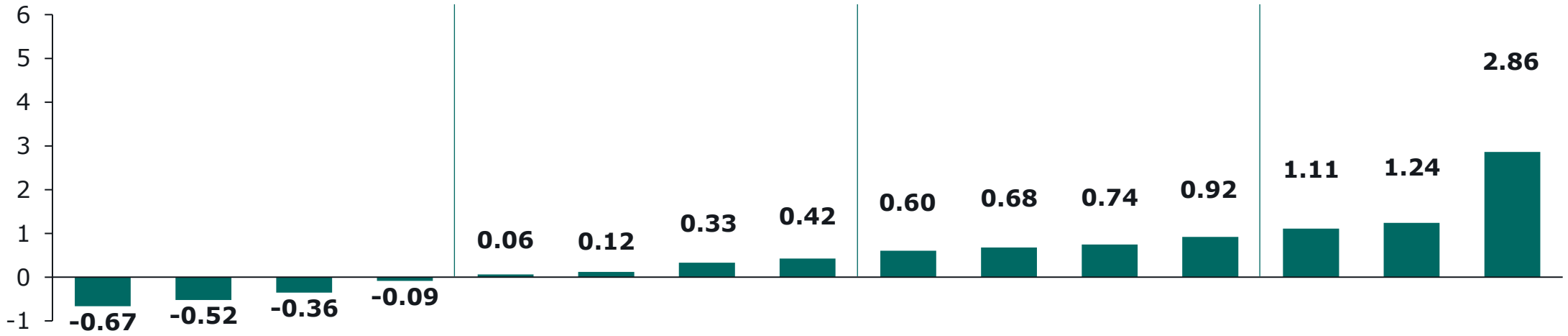
Overweight analysis: Actual vs Nominal GSM

Baseline

Actual vs Nominal GSM for CCM grades [11.2024-12.2024]

Total gross jumbo prod.:	194 447t	101 549t	137 212t	167 813t
W. avg. variability:	-0.4%	0.2%	0.7%	1.7%
Quartile:	Q1	Q2	Q3	Q4

% of Nominal GSM



Machine:	KS-06	WH-01	ZA-01	LC-02	BL-02	DU-01	KS-04	BL-03	AL-03	AB-01	DU-02	KS-03	VI-04	LC-01	RN-05
% Area sales:	82%	89%	65%	48%	93%	34%	88%	89%	70%	89%	70%	45%	40%	42%	65%
Nominal GSM:	96.2	137	118.4	133.9	130	133.2	132	115.1	158.2	107.3	129.7	137.3	163.4	131.5	104.5

GSM Actual vs Nominal

Overweight: Potential yearly savings (Maximum): Total 2.4 M$

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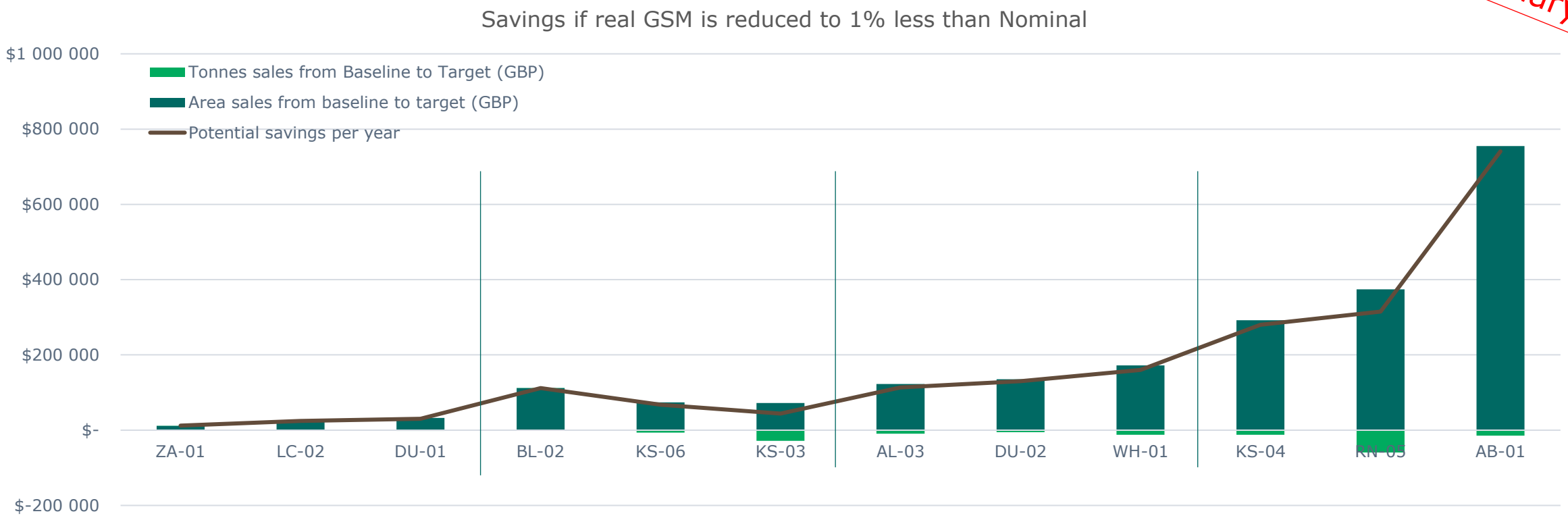

Machine:	ZA-01	LC-02	DU-01	BL-02	KS-06	KS-03	AL-03	DU-02	WH-01	KS-04	RN-05	AB-01
% Area sales:	65%	48%	34%	93%	82%	45%	70%	70%	89%	88%	65%	89%
Nominal GSM:	118.4	133.9	133.2	130	96.2	137.3	158.2	129.7	137	132	104.5	107.3

Financial values converted from GBP to USD – Exchange rate: 1,25; Tons=Short tons

[1] Viana and Lucca PM1 excluded from annual savings value of $2,59m

Strength Workstream: Savings Calculation for Underweighting

- Total Savings = [Area Sales Saving] + [Tonnes Sales Saving]
  - The overall production is split into two buckets based on sales type
  - Internal customers are typically area sales, while external customers are typically tonnes sales
  - Within a single grade, on a single machine, there can be both tonnes and area sales
  - Decisions on under or over weighting are applied to the whole grade
  - The calculation accounts for the portion of underweighted product sold by tonne, or vice versa, where there will be a negative saving

- Area Sales Saving =

  [Total Tonnes] * [% Area Sales] * [Cost per Tonne] * (-1 * [BW Ratio])
  - When selling by area, a lower grammage means fewer input costs for the same sales price, generating a saving
  - Cost per tonne includes fibre, steam and other costs
  - The basis weight ratio combined with the tonnes is effectively: “tonnes that would not have had to be produced to make the same area if running at the target basis weight instead of the baseline”
  - A positive value for Area Sales Saving is a saving, and negative is a loss
  - The BW Ratio will be negative when underweighting. It is multiplied by -1 to get a positive savings value when underweighting
  - It is assumed that there is no change in output rate (m/min)

- Tonnes Sales Saving =

  [Total Tonnes] * [% Tonnes Sales] * [Contribution] * [BW Ratio]
  - When selling by tonne, a higher grammage means more sales for the same area

- Contribution = [Sales Price per Tonne] - [Cost per Tonne]

- BW Ratio = ([Target] - [Baseline]) / [Baseline]

Example calculations overweight

- Aschaffenburg mill, Nov-Dec 2024, 6010085

- Tonnes produced: 10,573 t
- Area sales: 93%
- Average basis weight, scanner: 85.8 gsm
- Cost per tonne: 397 $/t
- Sales price: 459 $/t
- Target GSM = Nominal GSM = 85 gsm

- BW Ratio = (85 – 85.8) / 85 = -0.009
- Contribution = 459 – 397 = 61.6 $/t
- Tonnes sales saving = 10,573 * (1-93/100) * 61.6 * -0.009 = $-410
- Area sales saving = 10,573 * 93/100 * 397 * -1 * -0.009 = $35,133
- Total savings = 35133 – 410 = $34,723 (for two months)
- Savings per month = 34723/2 = $17,361

- If we had made this product at 85 gsm instead of 85.8 gsm, we assume that we would have produced about 1% fewer tonnes and sold the same area of paper.

- That 1% is about 100 tonnes which cost 400 $/t to produce, so not having to make it would save around $40,000 over the two months.

- The actual figure is a bit lower because not all of those tonnes are sold by area.